

A NOTE ON VARIANTS OF BUCHSTAB'S IDENTITY

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ABSTRACT. The author proves variants of Buchstab's identity on sieve functions, refining the previous work on new iteration rules of Brady. The main tool used in the proof is a special form of combinatorial identities related to the binomial coefficients. As a by-product, the author obtains better inequalities of $F_\kappa(s)$ and $f_\kappa(s)$ for dimensions $\kappa > 1$.

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1. INTRODUCTION

Let $\mathcal{A}$ be a finite set of integers, $\mathcal{A}_d = \{a : a \in \mathcal{A}, d \mid a\}$ and $S(\mathcal{A}, z) = \sum_{(a, \prod_{p < z} p) = 1}^{a \in \mathcal{A}}$. Suppose that κ, z, y are such that for every squarefree integer d , all of whose prime factors are less than z , we have

$$|\mathcal{A}_d| = \kappa^{\omega(d)} \frac{y}{d} + \text{error}, \quad (1)$$

where the error in (1) can be controlled on average. Suppose that $y = z^s$. Define $F_\kappa(s)$ and $f_\kappa(s)$ as the largest resp. smallest function such that the following bound holds for all sets $\mathcal{A}$ satisfying (1):

$$(1 + o(1))f_\kappa(s)y \prod_{p < z} \left(1 - \frac{\kappa}{p}\right) \leq S(\mathcal{A}, z) \leq (1 + o(1))F_\kappa(s)y \prod_{p < z} \left(1 - \frac{\kappa}{p}\right). \quad (2)$$

Selberg [3] has shown that $F_\kappa(s)$ and $f_\kappa(s)$ are continuous, monotone, and computable for $s > 1$, and that they tend to 1 exponentially as s goes to infinity.

When $\kappa \leq 1$, the optimal estimates for $F_\kappa(s)$ and $f_\kappa(s)$ arise from Buchstab's identity

$$S(\mathcal{A}, z) = S(\mathcal{A}, w) - \sum_{w \leq p < z} S(\mathcal{A}_p, p) \quad (3)$$

for $w \leq z$. Simply let $w = 2$, this becomes

$$S(\mathcal{A}, z) = |\mathcal{A}| - \sum_{p < z} S(\mathcal{A}_p, p). \quad (4)$$

This leads to the inequalities

$$s^\kappa F_\kappa(s) \leq s^\kappa - \kappa \int_{t > s} t^{\kappa-1} (f_\kappa(t-1) - 1) dt, \quad (5)$$

$$s^\kappa f_\kappa(s) \geq s^\kappa - \kappa \int_{t > s} t^{\kappa-1} (F_\kappa(t-1) - 1) dt. \quad (6)$$

Infinite iteration of these inequalities leads to the β -sieve.

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However, there are better estimates for $F_\kappa(s)$ and $f_\kappa(s)$ when $\kappa > 1$. Taking Selberg's upper bound sieve as a starting point and using similar iteration rules, Diamond, Halberstam and Richert [2] developed their DHR-sieve.

In 2017, Brady mentioned and proved lots of new sieve iteration rules in his PhD thesis. One of his simplest upper bound sieve is

$$S(\mathcal{A}, z) \leq S(\mathcal{A}, w) - \frac{2}{3} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) + \frac{1}{3} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w).$$

He proved this inequality using a combinatorial identity

$$1 - \frac{2}{3}n + \frac{1}{3} \binom{n}{2} = \left(1 - \frac{n}{2}\right) \left(1 - \frac{n}{3}\right) \geq 0. \quad (7)$$

Clearly, this leads to an inequality of $F_\kappa(s)$:

$$s^\kappa F_\kappa(s) \leq t^\kappa F_\kappa(t) - \frac{2}{3} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa f_\kappa(t(1-x_1))}{x_1} dx_1 + \frac{1}{3} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa F_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1. \quad (8)$$

In this note, we further develop his method and prove a series of generalized iteration rules.

2. UPPER BOUND ITERATION

We first prove a simple upper bound iteration, which is a direct generalization of [[1], Theorem 34].

Theorem 2.1. *For any odd positive integer k and $w \leq z$, we have*

$$\begin{aligned} S(\mathcal{A}, z) &\leq S(\mathcal{A}, w) - \frac{k-1}{k} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) + \frac{k-2}{k} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w) \\ &\quad - \frac{k-3}{k} \sum_{w \leq p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3}, w) + \cdots \\ &\quad - \frac{2}{k} \sum_{w \leq p_{k-2} < \cdots < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 \cdots p_{k-2}}, w) \\ &\quad + \frac{1}{k} \sum_{w \leq p_{k-1} < \cdots < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 \cdots p_{k-1}}, w). \end{aligned}$$

Proof. We follow the essential steps in the proof of [[1], Theorem 34]. Let $a \in \mathcal{A}$. If a has any prime factor below w , then both quantities are clearly zero. Assume that a has no prime factors below w and has exactly n prime factors between w and z . If $n = 0$ then both sides count a once. Thus we only need to show that for any integer $n \geq 1$ we have

$$0 \leq 1 - \frac{k-1}{k}n + \frac{k-2}{k} \binom{n}{2} - \frac{k-3}{k} \binom{n}{3} + \cdots + \frac{1}{k} \binom{n}{k-1}. \quad (9)$$

Note that we have the following identity

$$1 - \frac{k-1}{k}n + \frac{k-2}{k} \binom{n}{2} - \frac{k-3}{k} \binom{n}{3} + \cdots + \frac{1}{k} \binom{n}{k-1} = \left(1 - \frac{n}{2}\right) \left(1 - \frac{n}{3}\right) \cdots \left(1 - \frac{n}{k}\right) \quad (10)$$

and the right hand side of (10), which has even number of terms, is clearly ≥ 0 , Theorem 2.1 is proved. Note that [[1], Theorem 34] is just Theorem 2.1 with $k = 3$. $\square$

Corollary 2.2. *For any odd positive integer k and real $2 \leq s \leq t$, we have*

$$\begin{aligned} s^\kappa F_\kappa(s) &\leq t^\kappa F_\kappa(t) - \frac{k-1}{k} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa f_\kappa(t(1-x_1))}{x_1} dx_1 + \frac{k-2}{k} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa F_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1 \\ &\quad - \frac{k-3}{k} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \frac{t^\kappa f_\kappa(t(1-x_1-x_2-x_3))}{x_1 x_2 x_3} dx_3 dx_2 dx_1 + \cdots \end{aligned}$$

$$\begin{aligned}
& -\frac{2}{k} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \cdots \int_{\frac{1}{t}}^{x_{k-3}} \frac{t^\kappa f_\kappa(t(1-x_1-x_2-\cdots-x_{k-2}))}{x_1 x_2 \cdots x_{k-2}} dx_{k-2} \cdots dx_2 dx_1 \\
& + \frac{1}{k} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \cdots \int_{\frac{1}{t}}^{x_{k-2}} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-\cdots-x_{k-1}))}{x_1 x_2 \cdots x_{k-1}} dx_{k-1} \cdots dx_2 dx_1.
\end{aligned}$$

However, we can use more flexible parameters to get more variants of this iteration. Before stating the next result, we first define

$$\mathcal{U} = \{(m_1, m_2) : 0 < m_1 \leq m_2 \leq \lfloor m_1 \rfloor + 1 \text{ or } 0 < m_2 \leq m_1 \leq \lfloor m_2 \rfloor + 1\}. \quad (11)$$

Theorem 2.3. *For any m_1, m_2 such that $(m_1, m_2) \in \mathcal{U}$ and $w \leq z$, we have*

$$S(\mathcal{A}, z) \leq S(\mathcal{A}, w) - \frac{m_1 + m_2 - 1}{m_1 m_2} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) + \frac{2}{m_1 m_2} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w).$$

Proof. Again, we use the essentially same arguments as the proof of Theorem 2.1. Let $a \in \mathcal{A}$. If a has any prime factor below w , then both quantities are clearly zero. Assume that a has no prime factors below w and has exactly n prime factors between w and z . If $n = 0$ then both sides count a once. Thus we only need to show that for any integer $n \geq 1$ we have

$$0 \leq 1 - \frac{m_1 + m_2 - 1}{m_1 m_2} n + \frac{2}{m_1 m_2} \binom{n}{2}. \quad (12)$$

By the following identity

$$1 - \frac{m_1 + m_2 - 1}{m_1 m_2} n + \frac{2}{m_1 m_2} \binom{n}{2} = \left(1 - \frac{n}{m_1}\right) \left(1 - \frac{n}{m_2}\right) \quad (13)$$

and $(m_1, m_2) \in \mathcal{U}$, we know that the right-hand side of (13) is ≥ 0 for any positive integer n . Theorem 2.3 is thus proved. Note that [[1], Theorem 34] is just Theorem 2.3 with $m_1 = 2$ and $m_2 = 3$. $\square$

Corollary 2.4. *For any m_1, m_2 such that $(m_1, m_2) \in \mathcal{U}$ and real $2 \leq s \leq t$, we have*

$$\begin{aligned}
s^\kappa F_\kappa(s) & \leq t^\kappa F_\kappa(t) - \frac{m_1 + m_2 - 1}{m_1 m_2} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa f_\kappa(t(1-x_1))}{x_1} dx_1 \\
& + \frac{2}{m_1 m_2} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa F_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1.
\end{aligned}$$

Using the same method but with more parameters, we can get lots of upper bound iterations of this type. For the sake of simplicity, we write

$$M_k^r = \sum_{1 \leq i_1 < i_2 < \cdots < i_r \leq k} m_{i_1} m_{i_2} \cdots m_{i_r}.$$

Theorem 2.5. *For any m_1, m_2, m_3, m_4 such that $(m_1, m_2) \in \mathcal{U}$, $(m_3, m_4) \in \mathcal{U}$ and $w \leq z$, we have*

$$\begin{aligned}
S(\mathcal{A}, z) & \leq S(\mathcal{A}, w) - \frac{M_4^3 - M_4^2 + M_4^1 - 1}{m_1 m_2 m_3 m_4} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) + \frac{2(M_4^2 - 3M_4^1 + 7)}{m_1 m_2 m_3 m_4} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w) \\
& - \frac{6(M_4^1 - 6)}{m_1 m_2 m_3 m_4} \sum_{w \leq p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3}, w) + \frac{24}{m_1 m_2 m_3 m_4} \sum_{w \leq p_4 < p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3 p_4}, w).
\end{aligned}$$

Corollary 2.6. *For any m_1, m_2, m_3, m_4 such that $(m_1, m_2) \in \mathcal{U}$, $(m_3, m_4) \in \mathcal{U}$ and real $2 \leq s \leq t$, we have*

$$\begin{aligned}
s^\kappa F_\kappa(s) & \leq t^\kappa F_\kappa(t) - \frac{M_4^3 - M_4^2 + M_4^1 - 1}{m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa f_\kappa(t(1-x_1))}{x_1} dx_1 \\
& + \frac{2(M_4^2 - 3M_4^1 + 7)}{m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa F_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1
\end{aligned}$$

$$\begin{aligned}
& - \frac{6(M_4^1 - 6)}{m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \frac{t^\kappa f_\kappa(t(1-x_1-x_2-x_3))}{x_1 x_2 x_3} dx_3 dx_2 dx_1 \\
& + \frac{24}{m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \int_{\frac{1}{t}}^{x_3} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3-x_4))}{x_1 x_2 x_3 x_4} dx_4 dx_3 dx_2 dx_1.
\end{aligned}$$

Theorem 2.7. For any $m_1, m_2, m_3, m_4, m_5, m_6$ such that $(m_1, m_2) \in \mathcal{U}$, $(m_3, m_4) \in \mathcal{U}$, $(m_5, m_6) \in \mathcal{U}$ and $w \leq z$, we have

$$\begin{aligned}
S(\mathcal{A}, z) & \leq S(\mathcal{A}, w) - \frac{M_6^5 - M_6^4 + M_6^3 - M_6^2 + M_6^1 - 1}{m_1 m_2 m_3 m_4 m_5 m_6} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) \\
& + \frac{2(M_6^4 - 3M_6^3 + 7M_6^2 - 15M_6^1 + 31)}{m_1 m_2 m_3 m_4 m_5 m_6} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w) \\
& - \frac{6(M_6^3 - 6M_6^2 + 25M_6^1 - 90)}{m_1 m_2 m_3 m_4 m_5 m_6} \sum_{w \leq p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3}, w) \\
& + \frac{24(M_6^2 - 10M_6^1 + 65)}{m_1 m_2 m_3 m_4 m_5 m_6} \sum_{w \leq p_4 < p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3 p_4}, w) \\
& - \frac{120(M_6^1 - 15)}{m_1 m_2 m_3 m_4 m_5 m_6} \sum_{w \leq p_5 < p_4 < p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3 p_4 p_5}, w) \\
& + \frac{720}{m_1 m_2 m_3 m_4 m_5 m_6} \sum_{w \leq p_6 < p_5 < p_4 < p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3 p_4 p_5 p_6}, w).
\end{aligned}$$

Corollary 2.8. For any $m_1, m_2, m_3, m_4, m_5, m_6$ such that $(m_1, m_2) \in \mathcal{U}$, $(m_3, m_4) \in \mathcal{U}$, $(m_5, m_6) \in \mathcal{U}$ and real $2 \leq s \leq t$, we have

$$\begin{aligned}
s^\kappa F_\kappa(s) & \leq t^\kappa F_\kappa(t) - \frac{M_6^5 - M_6^4 + M_6^3 - M_6^2 + M_6^1 - 1}{m_1 m_2 m_3 m_4 m_5 m_6} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa f_\kappa(t(1-x_1))}{x_1} dx_1 \\
& + \frac{2(M_6^4 - 3M_6^3 + 7M_6^2 - 15M_6^1 + 31)}{m_1 m_2 m_3 m_4 m_5 m_6} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa F_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1 \\
& - \frac{6(M_6^3 - 6M_6^2 + 25M_6^1 - 90)}{m_1 m_2 m_3 m_4 m_5 m_6} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \frac{t^\kappa f_\kappa(t(1-x_1-x_2-x_3))}{x_1 x_2 x_3} dx_3 dx_2 dx_1 \\
& + \frac{24(M_6^2 - 10M_6^1 + 65)}{m_1 m_2 m_3 m_4 m_5 m_6} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \int_{\frac{1}{t}}^{x_3} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3-x_4))}{x_1 x_2 x_3 x_4} dx_4 dx_3 dx_2 dx_1 \\
& - \frac{120(M_6^1 - 15)}{m_1 m_2 m_3 m_4 m_5 m_6} \times \\
& \quad \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \int_{\frac{1}{t}}^{x_3} \int_{\frac{1}{t}}^{x_4} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3-x_4-x_5))}{x_1 x_2 x_3 x_4 x_5} dx_5 dx_4 dx_3 dx_2 dx_1 \\
& + \frac{720}{m_1 m_2 m_3 m_4 m_5 m_6} \times \\
& \quad \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \int_{\frac{1}{t}}^{x_3} \int_{\frac{1}{t}}^{x_4} \int_{\frac{1}{t}}^{x_5} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3-x_4-x_5-x_6))}{x_1 x_2 x_3 x_4 x_5 x_6} dx_6 dx_5 dx_4 dx_3 dx_2 dx_1.
\end{aligned}$$

Clearly, Theorems 2.3, 2.5 and 2.7 are only the first of an infinite series of inequalities.

3. LOWER BOUND ITERATION

In this section we shall use a similar method to prove corresponding lower bound iterations.

Theorem 3.1. For any $0 < m_0 \leq 1$ and m_1, m_2 such that $(m_1, m_2) \in \mathcal{U}$ and $w \leq z$, we have

$$\begin{aligned} S(\mathcal{A}, z) &\geq S(\mathcal{A}, w) - \frac{m_0 m_1 + m_0 m_2 + m_1 m_2 - m_0 - m_1 - m_2 + 1}{m_0 m_1 m_2} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) \\ &\quad + \frac{2(m_0 + m_1 + m_2 - 3)}{m_0 m_1 m_2} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w) \\ &\quad - \frac{6}{m_0 m_1 m_2} \sum_{w \leq p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3}, w). \end{aligned}$$

Proof. By the same arguments as in the proof of Theorem , we only need to show that for any integer $n \geq 1$ we have

$$0 \geq 1 - \frac{m_0 m_1 + m_0 m_2 + m_1 m_2 - m_0 - m_1 - m_2 + 1}{m_0 m_1 m_2} n + \frac{2(m_0 + m_1 + m_2 - 3)}{m_0 m_1 m_2} \binom{n}{2} - \frac{6}{m_0 m_1 m_2} \binom{n}{3}. \quad (14)$$

Here, we have the identity

$$\begin{aligned} &1 - \frac{m_0 m_1 + m_0 m_2 + m_1 m_2 - m_0 - m_1 - m_2 + 1}{m_0 m_1 m_2} n + \frac{2(m_0 + m_1 + m_2 - 3)}{m_0 m_1 m_2} \binom{n}{2} - \frac{6}{m_0 m_1 m_2} \binom{n}{3} \\ &= \left(1 - \frac{n}{m_0}\right) \left(1 - \frac{n}{m_1}\right) \left(1 - \frac{n}{m_2}\right). \end{aligned} \quad (15)$$

One can easily check that for $0 < m_0 \leq 1$ and m_1, m_2 such that $(m_1, m_2) \in \mathcal{U}$, the right-hand side is ≤ 0 for any positive integer n . Hence Theorem 3.1 is proved. $\square$

Corollary 3.2. For any $0 < m_0 \leq 1$ and m_1, m_2 such that $(m_1, m_2) \in \mathcal{U}$ and $3 \leq s \leq t$, we have

$$\begin{aligned} s^\kappa f_\kappa(s) &\geq t^\kappa f_\kappa(t) - \frac{m_0 m_1 + m_0 m_2 + m_1 m_2 - m_0 - m_1 - m_2 + 1}{m_0 m_1 m_2} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa F_\kappa(t(1-x_1))}{x_1} dx_1 \\ &\quad + \frac{2(m_0 + m_1 + m_2 - 3)}{m_0 m_1 m_2} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa f_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1 \\ &\quad - \frac{6}{m_0 m_1 m_2} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3))}{x_1 x_2 x_3} dx_3 dx_2 dx_1. \end{aligned}$$

Again, for the sake of simplicity, we write

$$N_k^r = \sum_{0 \leq i_1 < i_2 < \dots < i_r \leq k-1} m_{i_1} m_{i_2} \dots m_{i_r}.$$

Theorem 3.3. For any $0 < m_0 \leq 1$, m_1, m_2, m_3, m_4 such that $(m_1, m_2) \in \mathcal{U}$, $(m_3, m_4) \in \mathcal{U}$ and $w \leq z$, we have

$$\begin{aligned} S(\mathcal{A}, z) &\geq S(\mathcal{A}, w) - \frac{N_5^4 - N_5^3 + N_5^2 - N_5^1 + 1}{m_0 m_1 m_2 m_3 m_4} \sum_{w \leq p_1 < z} S(\mathcal{A}_{p_1}, w) \\ &\quad + \frac{2(N_5^3 - 3N_5^2 + 7N_5^1 - 15)}{m_0 m_1 m_2 m_3 m_4} \sum_{w \leq p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2}, w) \\ &\quad - \frac{6(N_5^2 - 6N_5^1 + 25)}{m_0 m_1 m_2 m_3 m_4} \sum_{w \leq p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3}, w) \\ &\quad + \frac{24(N_5^1 - 10)}{m_0 m_1 m_2 m_3 m_4} \sum_{w \leq p_4 < p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3 p_4}, w) \\ &\quad - \frac{120}{m_0 m_1 m_2 m_3 m_4} \sum_{w \leq p_5 < p_4 < p_3 < p_2 < p_1 < z} S(\mathcal{A}_{p_1 p_2 p_3 p_4 p_5}, w). \end{aligned}$$

Corollary 3.4. *For any $0 < m_0 \leq 1$, m_1, m_2, m_3, m_4 such that $(m_1, m_2) \in \mathcal{U}$, $(m_3, m_4) \in \mathcal{U}$ and $3 \leq s \leq t$, we have*

$$\begin{aligned}
s^\kappa f_\kappa(s) \geq & t^\kappa f_\kappa(t) - \frac{N_5^4 - N_5^3 + N_5^2 - N_5^1 + 1}{m_0 m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \frac{t^\kappa F_\kappa(t(1-x_1))}{x_1} dx_1 \\
& + \frac{2(N_5^3 - 3N_5^2 + 7N_5^1 - 15)}{m_0 m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \frac{t^\kappa f_\kappa(t(1-x_1-x_2))}{x_1 x_2} dx_2 dx_1 \\
& - \frac{6(N_5^2 - 6N_5^1 + 25)}{m_0 m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3))}{x_1 x_2 x_3} dx_3 dx_2 dx_1 \\
& + \frac{24(N_5^1 - 10)}{m_0 m_1 m_2 m_3 m_4} \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \int_{\frac{1}{t}}^{x_3} \frac{t^\kappa f_\kappa(t(1-x_1-x_2-x_3-x_4))}{x_1 x_2 x_3 x_4} dx_4 dx_3 dx_2 dx_1 \\
& - \frac{120}{m_0 m_1 m_2 m_3 m_4} \times \\
& \int_{\frac{1}{t}}^{\frac{1}{s}} \int_{\frac{1}{t}}^{x_1} \int_{\frac{1}{t}}^{x_2} \int_{\frac{1}{t}}^{x_3} \int_{\frac{1}{t}}^{x_4} \frac{t^\kappa F_\kappa(t(1-x_1-x_2-x_3-x_4-x_5))}{x_1 x_2 x_3 x_4 x_5} dx_5 dx_4 dx_3 dx_2 dx_1.
\end{aligned}$$

4. FURTHER PROSPECT

In this note, we only give some sieve inequalities and do not mention any possible application of these inequalities. In fact, these may be helpful in bounding the "sifting limits" β_κ for $\kappa > 1$. The bounds for β_κ are quite important in many high-dimensional sieve problems. We hope someone can accomplish this work.

There are many other iteration rules proved in Brady's thesis [1]. We state two of them in the rest of this note, and we hope someone can generalize them.

Theorem 4.1. ([1], Theorem 35]. *For any $w \leq z^2$, we have*

$$\begin{aligned}
S(\mathcal{A}, z) \geq & S\left(\mathcal{A}, w^{\frac{1}{2}}\right) - \sum_{w^{\frac{1}{2}} \leq p_1 < z} S\left(\mathcal{A}_{p_1}, \frac{w}{p_1}\right) + \frac{5}{6} \sum_{\frac{w}{p_1} \leq p_2 < p_1 < z} S\left(\mathcal{A}_{p_1 p_2}, \frac{w}{p_1}\right) \\
& - \frac{2}{3} \sum_{\frac{w}{p_1} \leq p_3 < p_2 < p_1 < z} S\left(\mathcal{A}_{p_1 p_2 p_3}, \frac{w}{p_1}\right) - \frac{1}{2} \sum_{\frac{w}{p_2} \leq p_3 < p_2 < p_1 < z} S\left(\mathcal{A}_{p_1 p_2 p_3}, \frac{w}{p_1}\right).
\end{aligned}$$

Theorem 4.2. ([1], Theorem 42]. *If every element of $\mathcal{A}$ has size at most $y^{\frac{13}{12}}$ and $z^{\frac{12}{5}} < y < z^{\frac{5}{2}}$, we have*

$$\begin{aligned}
S(\mathcal{A}, z) \leq & S\left(\mathcal{A}, \frac{y}{z^2}\right) - \frac{4}{5} \sum_{\frac{y}{z^2} \leq p_1 < \frac{z^3}{y}} S\left(\mathcal{A}_{p_1}, \frac{y}{z^2}\right) - \frac{2}{3} \sum_{\frac{z^3}{y} \leq p_1 < \frac{y^2}{z^4}} S\left(\mathcal{A}_{p_1}, \frac{y}{z^2}\right) - \frac{8}{15} \sum_{\frac{y^2}{z^4} \leq p_1 < z} S\left(\mathcal{A}_{p_1}, \frac{y}{z^2}\right) \\
& + \frac{3}{5} \sum_{\frac{y}{z^2} \leq p_2 < p_1 < \frac{z^3}{y}} S\left(\mathcal{A}_{p_1 p_2}, \frac{y}{z^2}\right) + \frac{7}{15} \sum_{\frac{y}{z^2} \leq p_2 < \frac{z^3}{y} \leq p_1 < \frac{y^2}{z^4}} S\left(\mathcal{A}_{p_1 p_2}, \frac{y}{z^2}\right) \\
& + \frac{1}{3} \sum_{\frac{y}{z^2} \leq p_2 < \frac{z^3}{y} < \frac{y^2}{z^4} \leq p_1 < z} S\left(\mathcal{A}_{p_1 p_2}, \frac{y}{z^2}\right) + \frac{1}{3} \sum_{\frac{z^3}{y} \leq p_2 < p_1 < \frac{y^2}{z^4}} S\left(\mathcal{A}_{p_1 p_2}, \frac{y}{z^2}\right) \\
& + \frac{4}{15} \sum_{\frac{z^3}{y} \leq p_2 < \frac{y^2}{z^4} \leq p_1 < z} S\left(\mathcal{A}_{p_1 p_2}, \frac{y}{z^2}\right) + \frac{1}{5} \sum_{\frac{y^2}{z^4} \leq p_2 < p_1 < z} S\left(\mathcal{A}_{p_1 p_2}, \frac{y}{z^2}\right) \\
& - \frac{2}{5} \sum_{\substack{\frac{y}{z^2} \leq p_3 < p_2 < p_1 < \frac{z^3}{y} \\ p_1 p_2 p_3^2 < z^2}} S\left(\mathcal{A}_{p_1 p_2 p_3}, \frac{y}{z^2}\right) \\
& - \frac{4}{15} \sum_{\frac{y}{z^2} \leq p_3 < p_2 < \frac{z^3}{y} \leq p_1 < \frac{y^2}{z^4}} \left(1 - \frac{3 \log(p_2 p_3)}{8 \log(y/p_1)}\right) S\left(\mathcal{A}_{p_1 p_2 p_3}, \frac{y}{z^2}\right)
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{5} \sum_{\substack{\frac{y}{z^2} \leq p_4 < p_3 < p_2 < p_1 < \frac{z^3}{y} \\ p_1 p_2 p_3^2 < z^2}} S\left(\mathcal{A}_{p_1 p_2 p_3 p_4}, \frac{y}{z^2}\right) \\
& + \frac{1}{10} \sum_{\substack{\frac{y}{z^2} \leq p_4 < p_3 < p_2 < \frac{z^3}{y} \leq p_1 < \frac{y^2}{z^4}}} \left(1 - \frac{\log(p_2 p_3 p_4)}{\log(y/p_1)}\right) S\left(\mathcal{A}_{p_1 p_2 p_3 p_4}, \frac{y}{z^2}\right).
\end{aligned}$$

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